BEACON

Guiding you through Campus Fog

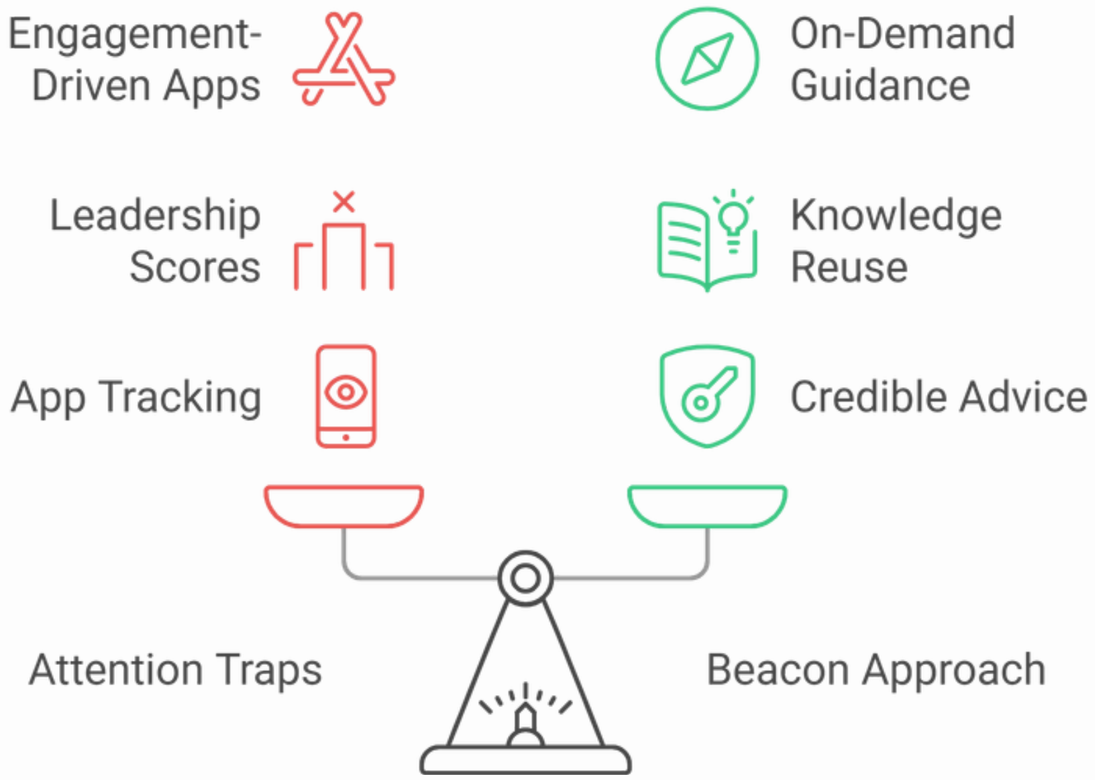
Team 11 - Laadle

Theme A - Attention Economy on Campus

Problem Statement

Students are overwhelmed by constant stimuli, causing fragmented focus and misaligned priorities

Designing Beyond Gamification



Students don't need more engagement loops - they need clearer guidance

Principles Behind BEACON

Less noise. Better guidance. Trusted mentorship.

Intent-Driven Interaction

- No endless feeds
- No gamified streaks
- Minimal notifications
- Focused usage

Knowledge Over Noise

- Structured guidance
- Reusable insights
- Faster answers
- Reduced repetition

Trust-Based Mentorship

- Credibility signals
- Hidden scoring
- No leaderboards
- Verified advice

Context-Aware Guidance Engine

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Untitled-1

Domain: Branch Change
Intent: Urgency
Student State: 2nd year, CGPA 7.1

Conversation:
"I need to switch branch this semester. What are my chances?"
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Untitled-1

Domain: Placement
Intent: Strategy Planning
Student State: 3rd year, aiming for product roles

Conversation:
"How should I prepare for product-based company placements?"
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Different Domain,

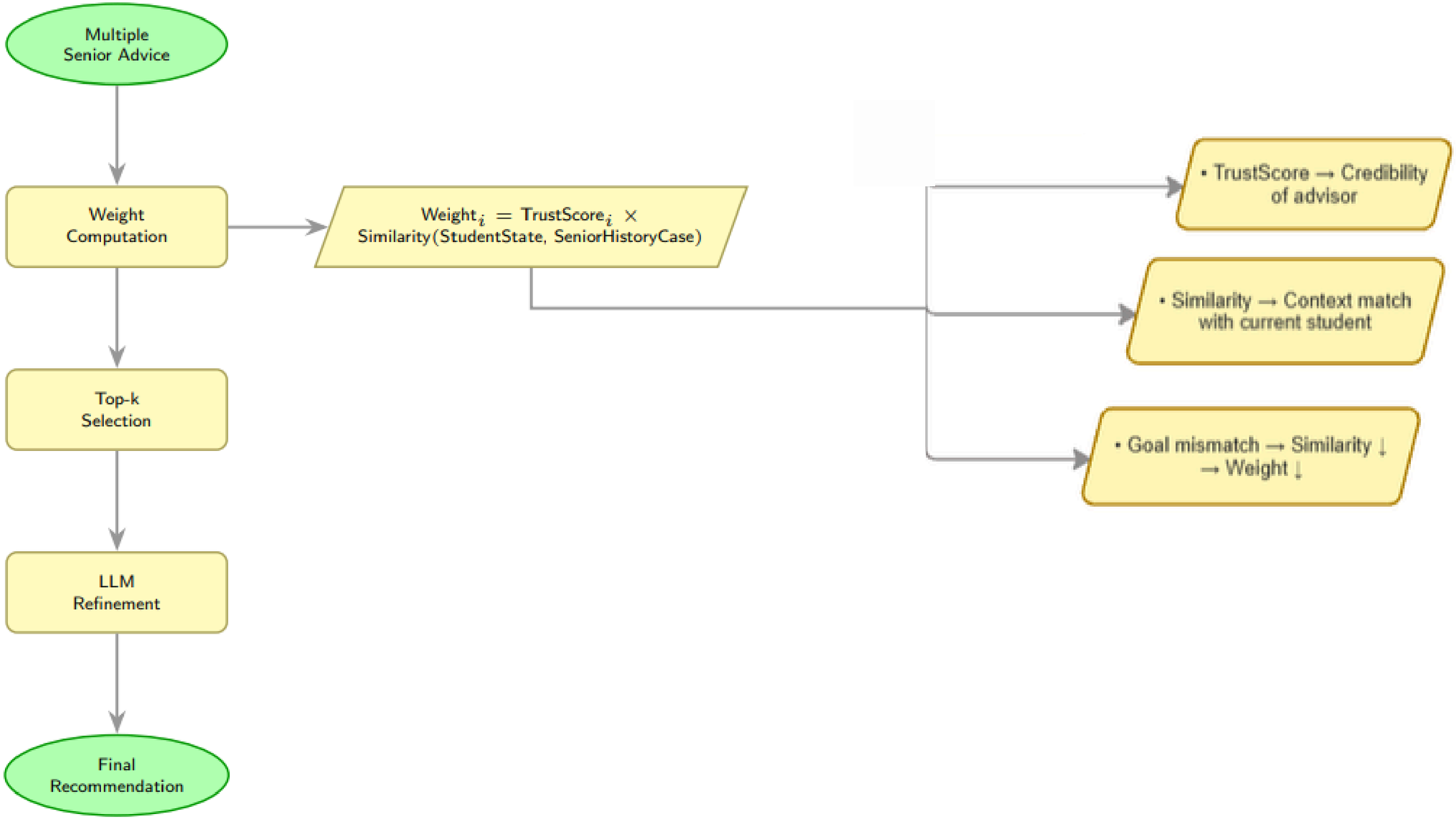
Different Intent.

Features

- Senior-Junior Mentorship
  Goal-aligned guidance from verified seniors

- Peer-to-Peer Collaboration
  Structured collaboration with like-minded peers

Weighted Advice Aggregation Mechanism



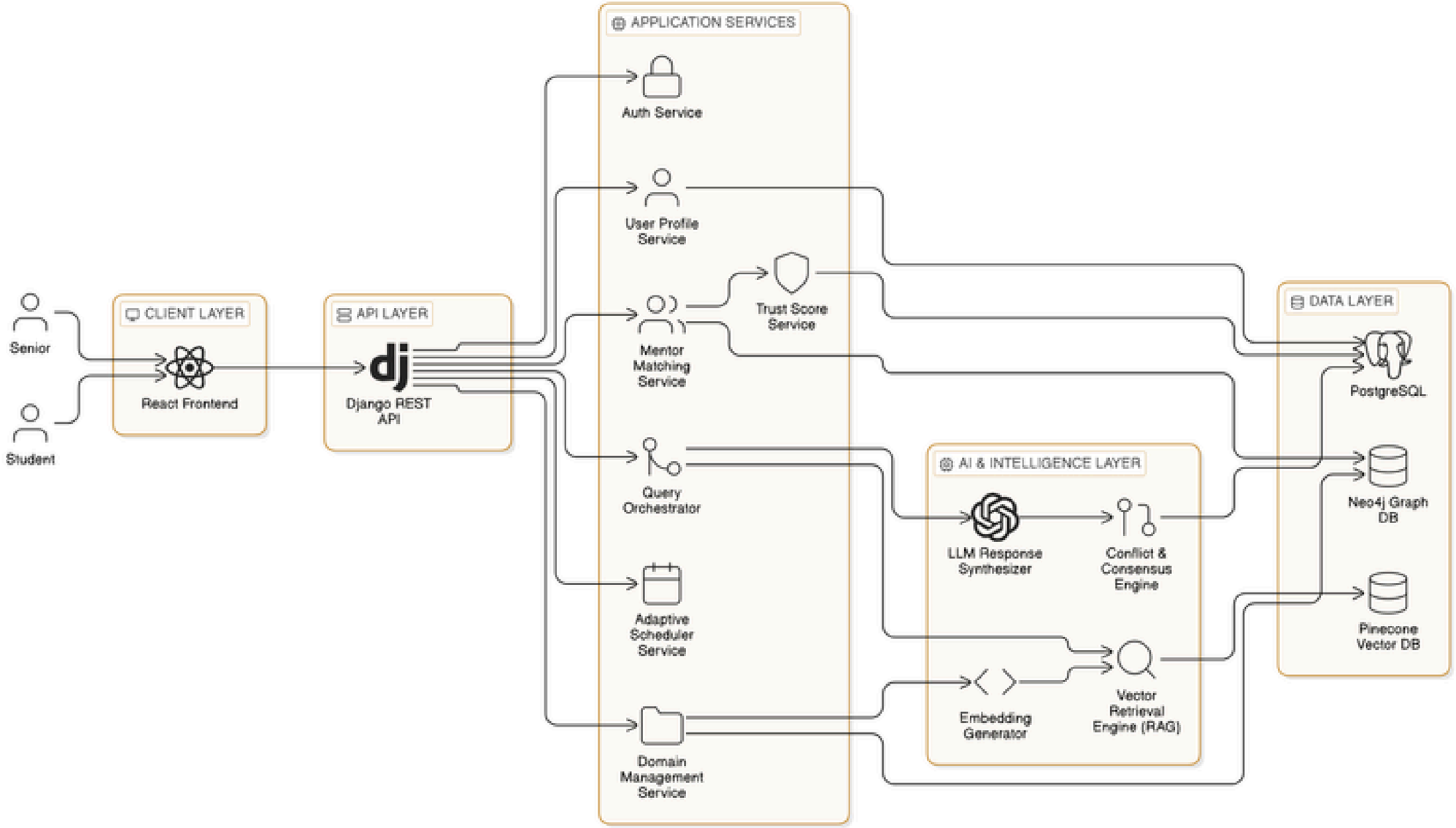
Multi-Database Strategy

PostgreSQL
Relational storage for structured application data

Neo4j
Represents mentor expertise relationships

Pinecone
Enables semantic search using embeddings

System Architecture



Trust Score Computation

Student Feedback Score

Student Satisfaction • Exponential Moving Average

Follow-Through Success Rate

Advice Outcome Rate • Laplace Smoothing

Consistency Score

Reply Speed Reliability • Exponential Decay

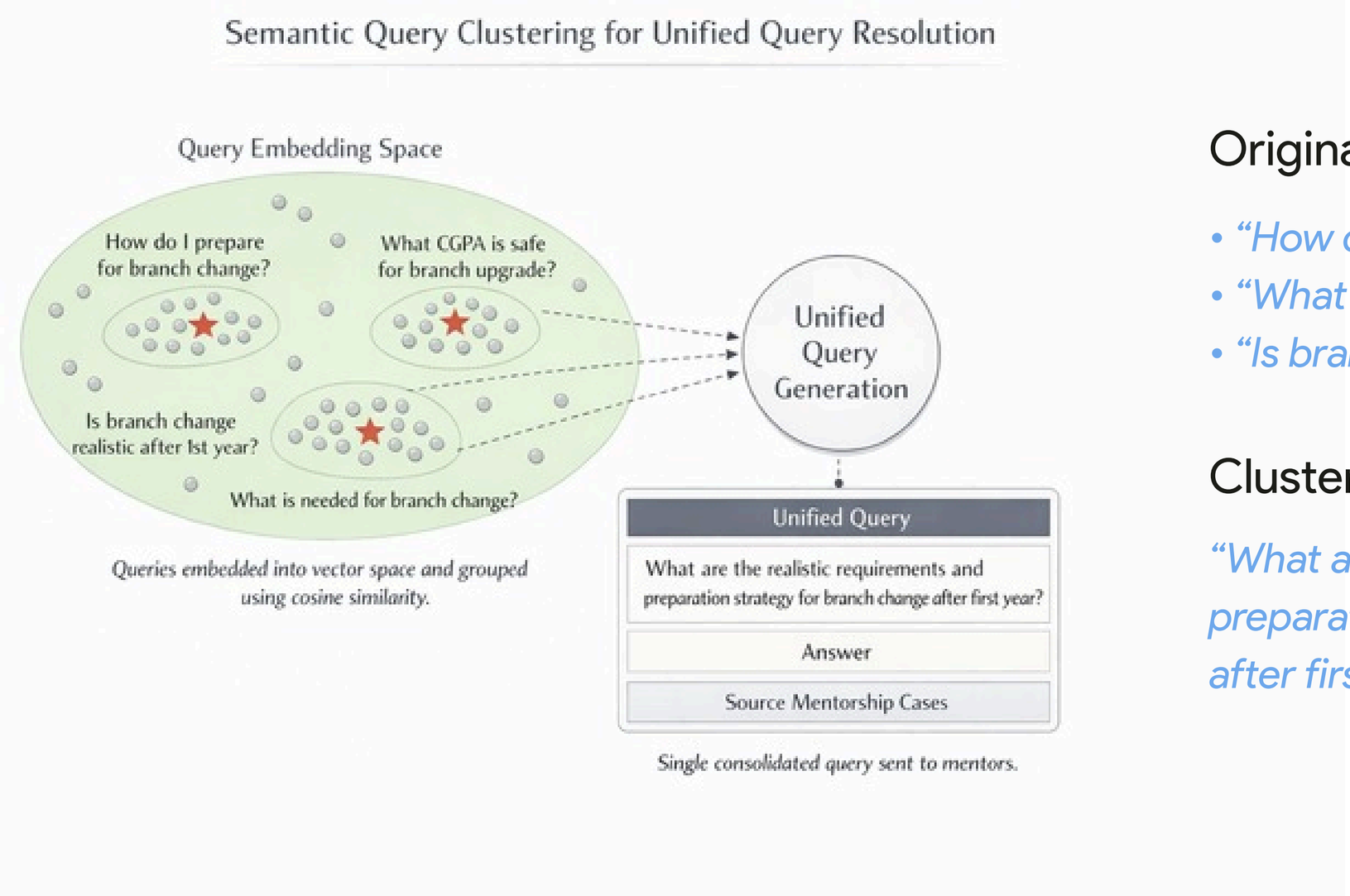
Alignment Score

Consensus Agreement • Majority Penalty

Verified Achievements

Expertise Relevance • Cosine Similarity

RAG & Unified Query Resolution



Original Queries:

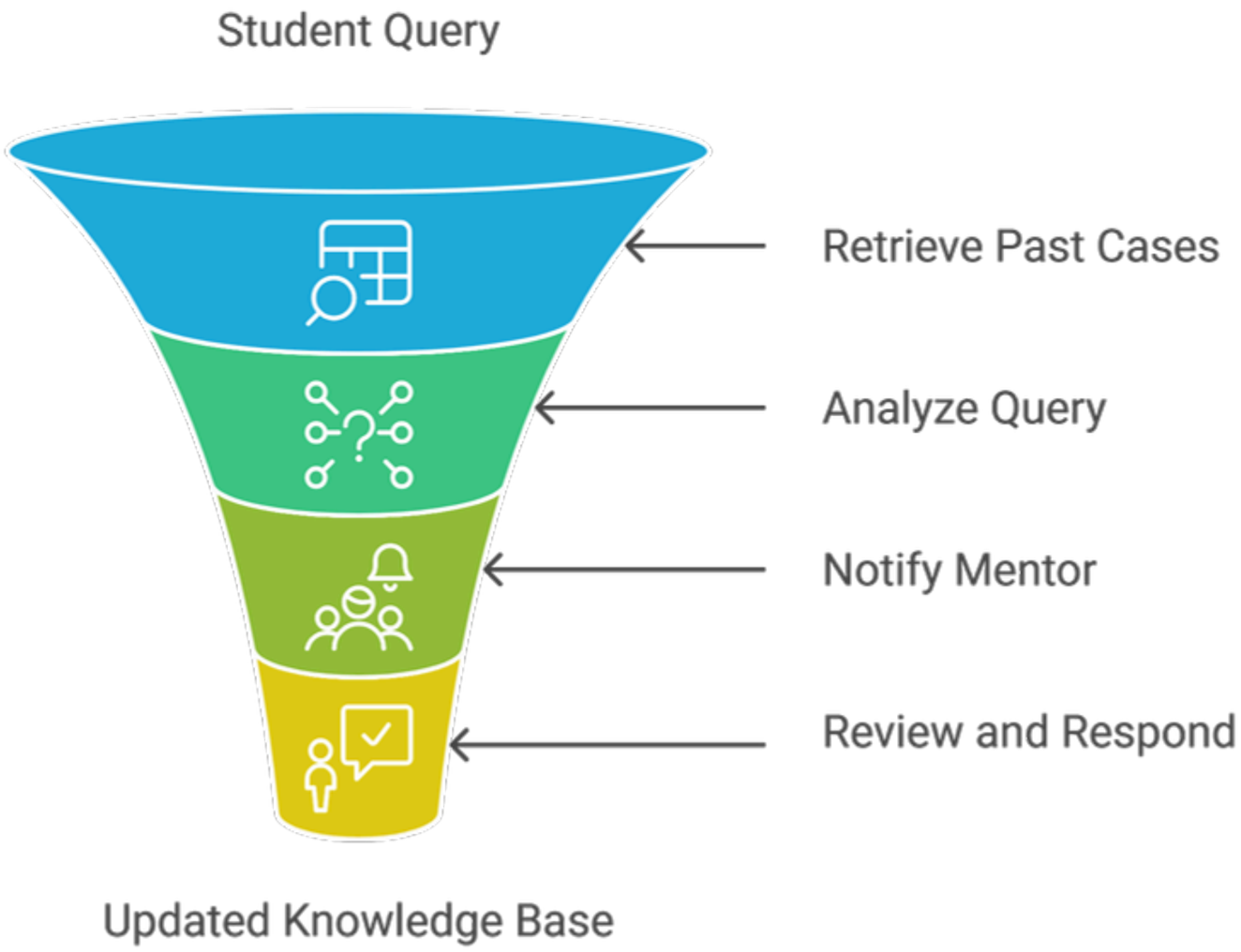
- *"How do I prepare for branch change?"*
- *"What CGPA is safe for branch upgrade?"*
- *"Is branch change realistic after 1st year?"*

Clustered Query to Senior:

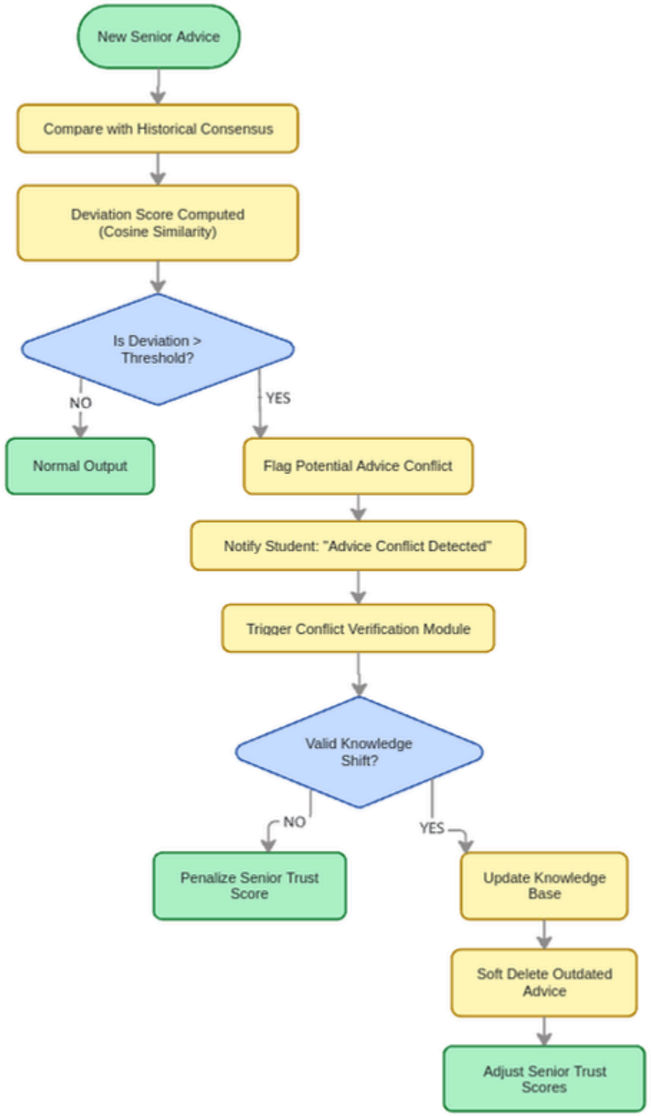
"What are the realistic requirements and preparation strategy for branch change after first year?"

Predictive FAQ

- Instant Data-Backed First Look
- Transparent Automation
- Predictive Follow-Up Generation
- Loop Reduction

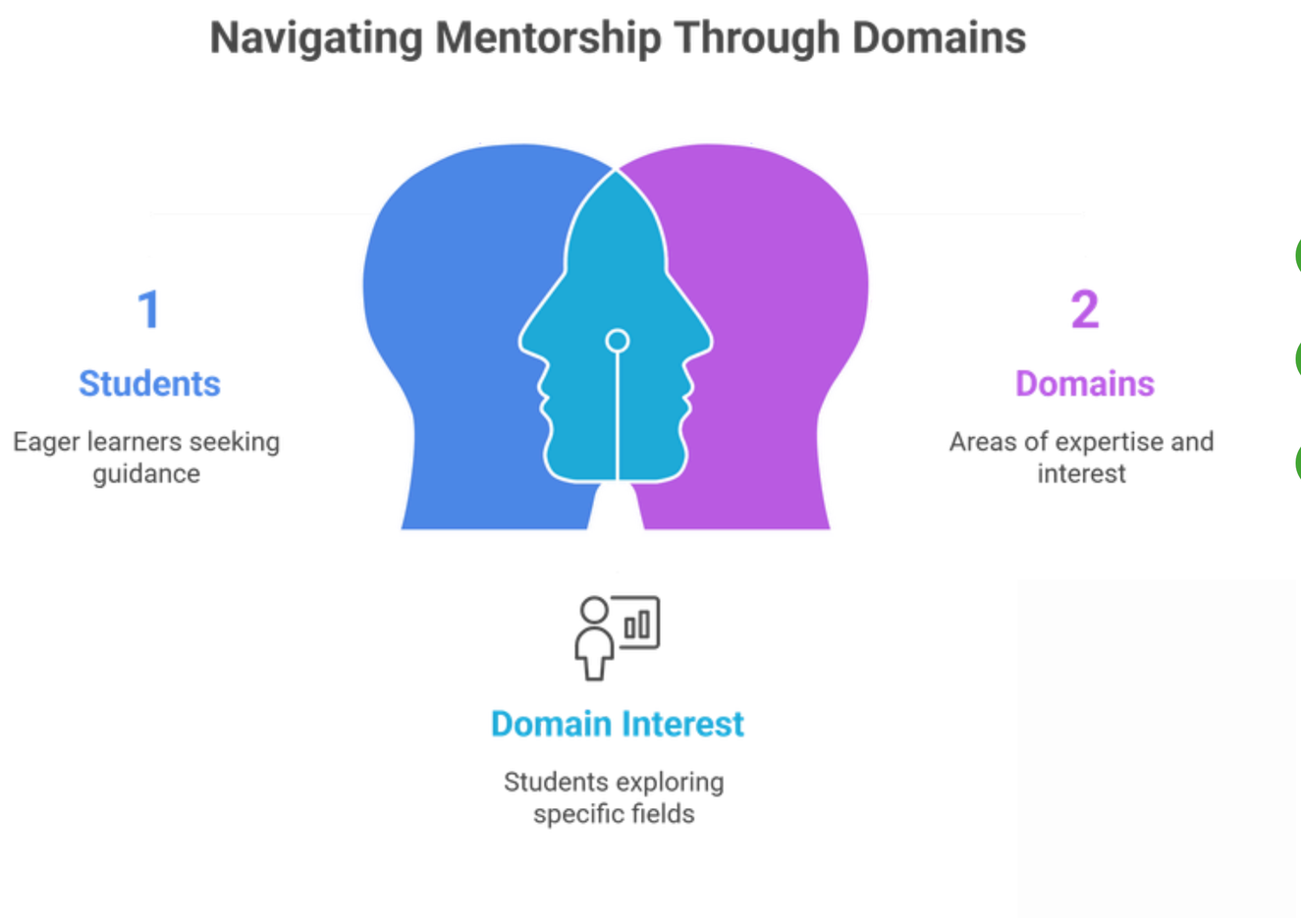


Conflict & Anomaly Detection



- Detects deviation from historical consensus
- Requests verification if deviation crosses threshold
- Soft-deletes outdated guidance upon confirmation
- Dynamically updates senior trust scores

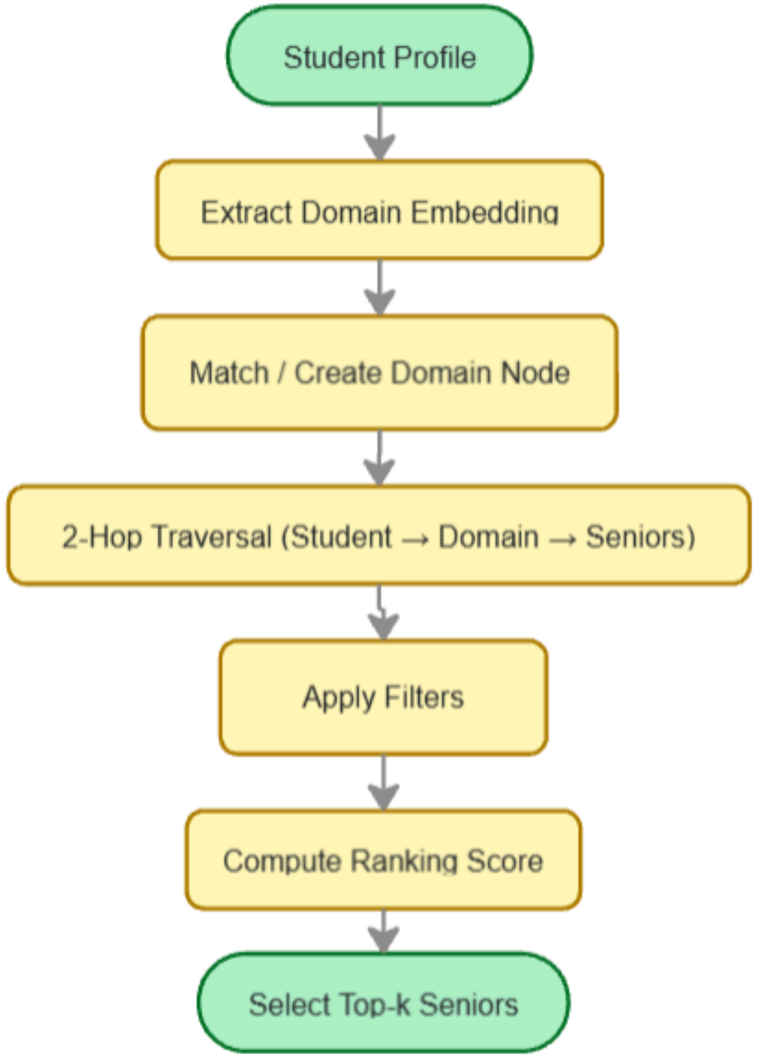
Intelligent Graph Matching



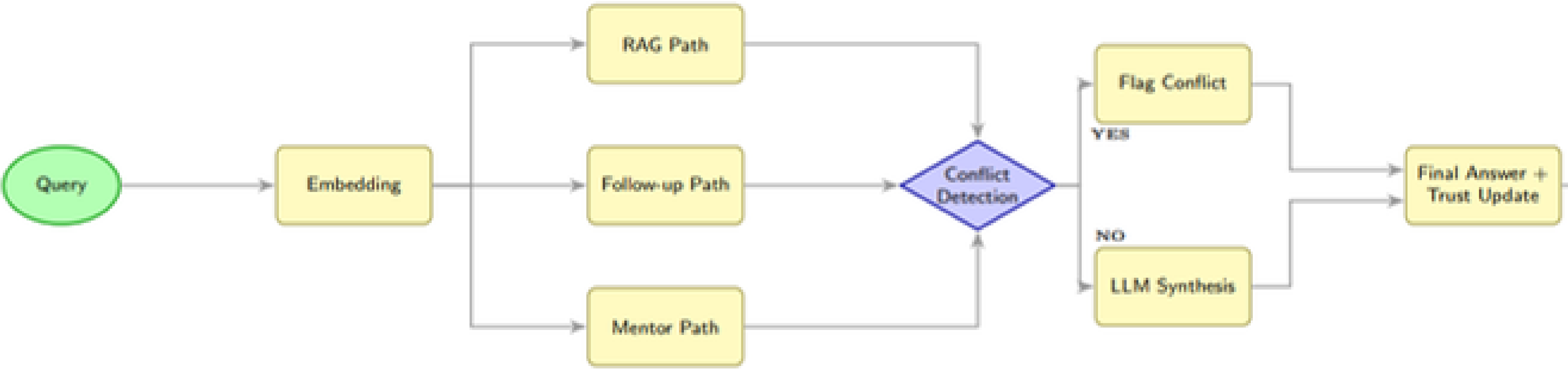
- Domain-Centric Matching
- Semantic Domain Normalization
- Multi-Relationship Support

Matching Algorithm

- 2-Hop Graph Traversal
- Multi-Factor Filtering
- Ranking Formula
- Top-k Selection



ML Pipeline


Query
Embedding
RAG Path
Follow-up Path
Mentor Path
Conflict Detection
YES
NO
Flag Conflict
LLM Synthesis
Final Answer + Trust Update

Why Not Just Use Alternate Options?

Capabilities	WhatsApp Groups	RAG Bot on WhatsApp	Graph Mentorship (Neo4j)	BEACON (Ours)
Smart Mentor Discovery	✗	✗	✓	✓
Instant Answer	✗	✓	✗	✓
Knowledge Memory	✗	✓	✓	✓
Advice Conflict Handling	✗	✗	✗	✓
Trust Based Ranking	✗	✗	✗	✓

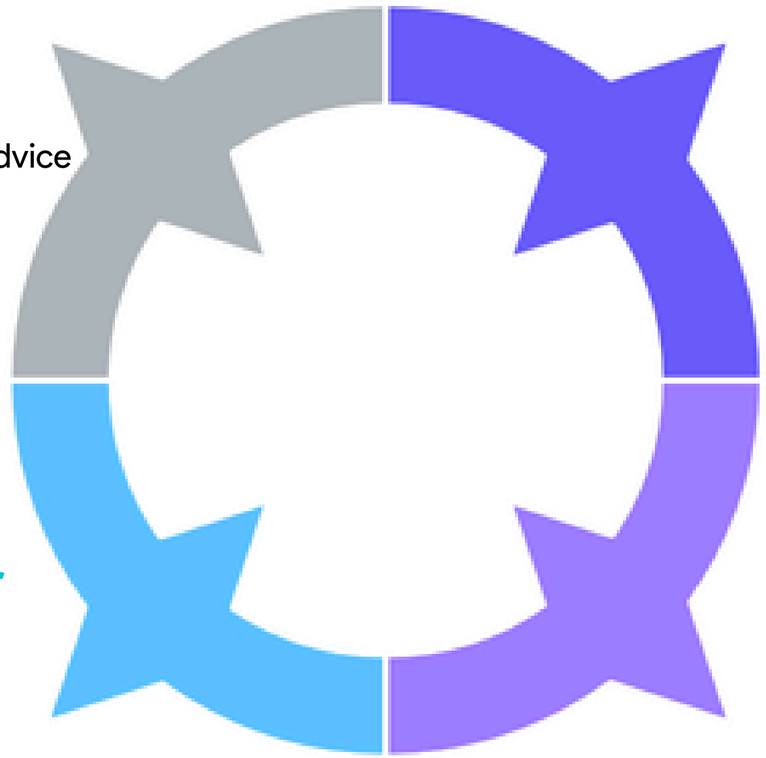
Future Roadmap & Monetization

Incentivization

Mentors earn XP or BEACON tokens when their advice is highly rated.

Corporate Bounties

Companies can sponsor specific knowledge domains (e.g., System Design, React). Outstanding answers to sponsored queries can earn referrals or monetary rewards.



Super Mentor Economy

Mentors with very high Trust Scores unlock elite “Super Mentor” status. They can offer paid 1-on-1 consultations and access premium mentoring opportunities.

Zero-Friction Mentoring

BEACON integrates with tools like Discord or Slack so mentors receive questions directly.They can answer instantly without opening the BEACON platform.

Thank You

Scalability